Problem 4 - Smart Kitchen Assistant

Week 7 | Category B Practice | Java solutions

Problem

KitchenTool has a private speedLevel accessible with getters/setters, validates values 1 through 5, and Blender implements Washable.

Complete Java Code

```
abstract class KitchenTool {
    private int speedLevel = 1;

    public abstract String prepare();

    public int getSpeedLevel() {
        return speedLevel;
    }

    public void setSpeedLevel(int speedLevel) {
        if (speedLevel < 1 || speedLevel > 5) {
            throw new IllegalArgumentException("speedLevel must be between 1 and 5");
        }
        this.speedLevel = speedLevel;
    }
}

interface Washable {
    String clean();
}

class Blender extends KitchenTool implements Washable {
    public Blender() { super(); }

    @Override
    public String prepare() {
        return "Blending at speed " + getSpeedLevel();
    }

    @Override
    public String clean() {
        return "Blender rinsed and dried";
    }
}

public class Main {
    public static void main(String[] args) {
        Blender b = new Blender();
        b.setSpeedLevel(3);
        System.out.println(b.getSpeedLevel());
        try {
            b.setSpeedLevel(9);
        } catch (IllegalArgumentException ex) {
            System.out.println("rejected, speed level stays " + b.getSpeedLevel());
        }
        System.out.println(b.prepare());
        System.out.println(b.clean());
    }
}
```

Sample Output

```
3
rejected, speed level stays 3
Blending at speed 3
Blender rinsed and dried
```

How it works

speedLevel is private. The setter validates before assignment and throws an exception for an invalid value, so an existing valid value is unchanged. Blender provides both food-preparation and cleaning behaviors.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.